REAL ANALYSIS II · SEMESTER 5 · MODULE 4

Answer key: Sequences of Functions

Formal solutions to all 40 questions of the model paper.

BEFORE YOU READ AN ANSWER

Each answer starts with **Know first**: topic codes from File 1. If one feels new, read it first, try the question yourself, then read the solution.

Grey "Marks" notes show a typical split (a guide, not an official scheme): statement · set-up · key step · conclusion.

Section A · Short answers

3 marks each.

1 PYQ MTS-25 Q9

Uniform norm; $\|\sin\|_{\mathbb{R}}$.

KNOW FIRST

T5 Bounded functions and the uniform norm P3 Supremum (least upper bound)

Let $A \subseteq \mathbb{R}$ and $\varphi : A \to \mathbb{R}$. φ is **bounded on A** if $\varphi(A)$ is a bounded set. For bounded φ the **uniform norm** of φ on A is

$$\|\varphi\|_A := \sup\{|\varphi(x)| : x \in A\}.$$

It satisfies $\|\varphi\|_A \leq \varepsilon \iff |\varphi(x)| \leq \varepsilon$ for all $x \in A$.

Since $|\sin x| \leq 1$ for all x and $\sin(\pi/2) = 1$, $\|\sin\|_{\mathbb{R}} = 1$.

Marks: bounded 1 · definition of norm 1 · example 1.

2 PYQ MTS-22 Q8

Define pointwise convergence.

KNOW FIRST

T1 Pointwise convergence P1 Limits of sequences

Let (f_n) be a sequence of functions on $A \subseteq \mathbb{R}$, $A_0 \subseteq A$ and $f : A_0 \to \mathbb{R}$. (f_n) **converges (pointwise) on A_0 to f** if for each $x \in A_0$ the sequence of numbers $(f_n(x))$ converges to $f(x)$. Equivalently: for each $\varepsilon > 0$ and each $x \in A_0$ there is $K(\varepsilon, x) \in \mathbb{N}$ with $|f_n(x) - f(x)| < \varepsilon$ for all $n \geq$

$K(\varepsilon, x)$. f is then unique (a sequence has at most one limit).

Marks: definition 2 · ε–K form or uniqueness 1.

3 PYQ MTS-26 Q9

Define uniform convergence; compare.

KNOW FIRST

T3 Uniform convergence M3 The order of quantifiers (a game)

(f_n) **converges uniformly on** A_0 to $f : A_0 \to \mathbb{R}$ if for each $\varepsilon > 0$ there is $K(\varepsilon) \in \mathbb{N}$, depending on ε but **not on x**, such that $n \geq K(\varepsilon)$ implies $|f_n(x) - f(x)| < \varepsilon$ for **all** $x \in A_0$.

Difference: in pointwise convergence K may depend on both ε and x ($\forall\varepsilon\,\forall x\,\exists K$); in uniform convergence one K serves every x ($\forall\varepsilon\,\exists K\,\forall x$). Uniform ⇒ pointwise, not conversely (e.g. x/n on $\mathbb{R}$).

Marks: definition 2 · difference with example 1.

4 PYQ MTS-25 Q10

$\lim e^{-nx}$, $x \geq 0$.

KNOW FIRST

T2 Finding the limit function P1 Limits of sequences

If $x = 0$: $e^{-n\cdot 0} = 1$ for all n, so the limit is 1. If $x > 0$: $e^{-nx} = (e^{-x})^n$ with $0 < e^{-x} < 1$, so $(e^{-x})^n \to 0$.

ANSWER $\lim e^{-nx} = 1$ **if** $x = 0$, **and** $= 0$ **if** $x > 0$.

Marks: x = 0 case 1 · x > 0 case 1 · limit function stated 1.

5 PYQ MTS-26 Q10

$\lim \dfrac{nx}{1+nx}$, $x \geq 0$.

KNOW FIRST

T2 Finding the limit function P2 Algebra of limits and the squeeze

x = 0: every term is 0, limit 0. x > 0: dividing by nx , $\dfrac{nx}{1+nx} = \dfrac{1}{\frac{1}{nx}+1} \to \dfrac{1}{0+1} = 1$.

ANSWER $f(0) = 0$, $f(x) = 1$ **for** $x > 0$.

Note: each f_n is continuous on $[0, \infty)$, the limit jumps at 0, so the convergence is not uniform there (T14).

Marks: two cases 2 · answer 1.

6 PYQ MAT-22 Q8

$\lim \dfrac{x^2+nx}{n}$.

KNOW FIRST

T2 Finding the limit function P2 Algebra of limits and the squeeze

$\dfrac{x^2+nx}{n} = \dfrac{x^2}{n} + x$. For fixed x, $x^2/n \to 0$, so the limit is x.

ANSWER $\lim \dfrac{x^2+nx}{n} = x$ **for every** $x \in \mathbb{R}$.

Marks: rewrite 1 · limit 2.

7 PYQ MAT-21 Q10

$\lim \dfrac{\cos(nx+n)}{n}$.

KNOW FIRST

T2 Finding the limit function P2 Algebra of limits and the squeeze

$|\cos(nx+n)| \le 1$, so $0 \le \left|\dfrac{\cos(nx+n)}{n}\right| \le \dfrac{1}{n} \to 0$. By the squeeze theorem the limit is 0.

ANSWER 0, for every $x \in \mathbb{R}$.

Marks: bound 1 · squeeze 1 · answer 1.

8 PYQ MTS-22 MCQ 12 · adapted

$F_n = \dfrac{1}{n}\sin(nx+n) \to 0$ **uniformly on $\mathbb{R}$; find $\|F_n\|$.**

KNOW FIRST

T8 The shortcut: a bound free of x T6 The norm test (Lemma 8.1.8)

For every $x \in \mathbb{R}$: $|F_n(x) - 0| \le \dfrac{1}{n}$. The bound does not depend on x, so $\|F_n\|_{\mathbb{R}} \le \dfrac{1}{n}$. It is attained: at $x = \dfrac{\pi}{2n} - 1$, $nx + n = \pi/2$ and $F_n(x) = \dfrac{1}{n}$. Hence $\|F_n\|_{\mathbb{R}} = \dfrac{1}{n} \to 0$, so $F_n \to 0$ uniformly on $\mathbb{R}$ (Lemma 8.1.8). (MCQ answer: limit 0, option B.)

ANSWER $\|F_n\|_{\mathbb{R}} = 1/n \to 0$**: uniform.**

Marks: bound 1 · norm 1 · conclusion 1.

9 PYQ MTS-22 Q9

Uniform convergence of x^n on (−1, 1].

KNOW FIRST

T2 Finding the limit function T4 Proving "NOT uniform" M5 Negating a statement

Pointwise limit: $g(x) = 0$ for $-1 < x < 1$, $g(1) = 1$. Take $\varepsilon_0 = \frac{1}{2}$, $n_k = k$, $x_k = (1/2)^{1/k} \in (0, 1)$. Then $|x_k^k - g(x_k)| = \frac{1}{2} - 0 = \frac{1}{2} \geq \varepsilon_0$ for every k. By Lemma 8.1.5 the convergence is **not uniform** on (−1, 1].

ANSWER **Converges pointwise to g; not uniformly.**

Marks: limit 1 · choice of x_k, ε_0 1 · conclusion 1.

10 PYQ MTS-22 Q10

$h_n = 2nxe^{-nx^2}$: $\lim \int_0^1 h_n \neq \int_0^1 h$.

KNOW FIRST

T12 Limits and integrals P6 Integrals you can compute

$H_n(x) = -e^{-nx^2}$ satisfies $H_n'(x) = 2nx\, e^{-nx^2} = h_n(x)$. By the Fundamental Theorem, $\int_0^1 h_n = H_n(1) - H_n(0) = -e^{-n} + 1 \to 1$. But $\int_0^1 h = 0$. Since $1 \neq 0$, $\lim \int_0^1 h_n \neq \int_0^1 h$. (Moreover $h_n(x) \to 0$ for every x, so h is the pointwise limit: for x > 0, $0 < h_n(x) < 4/(nx^3)$.)

ANSWER $\lim \int_0^1 h_n = 1 \neq 0 = \int_0^1 h$.

Marks: antiderivative 1 · value $1 - e^{-n}$ 1 · comparison 1.

11 PYQ MTS-22 MCQ 11 · adapted

x^n: on which set is the limit continuous?

KNOW FIRST

T2 Finding the limit function T11 Pointwise limits can break continuity

(−1, 1] limit 0 on (−1, 1), 1 at 1: jumps at 1, not continuous

(−2, 1] no limit for $x \le -1$: does not converge there

(0, 1] limit jumps at 1: not continuous

(−1, 1) limit 0: continuous

ANSWER **Only on (−1, 1) (option D).**

Marks: limit function 1 · checking sets 1 · answer 1.

12 PYQ MTS-22 MCQ 13 · adapted

$f_n \to f$ uniformly on $\mathbb{R}$. $2f_n$? f_n^2?

KNOW FIRST

T15 Sums and products T6 The norm test (Lemma 8.1.8)

$(2f_n)$: yes, $|2f_n(x) - 2f(x)| = 2|f_n(x) - f(x)|$, so $\|2f_n - 2f\| = 2\|f_n - f\| \to 0$.

(f_n^2): not necessarily. $f_n(x) = x + \dfrac{1}{n} \to x$ uniformly on $\mathbb{R}$ (norm $1/n$), but $f_n^2(x) - x^2 = \dfrac{2x}{n} + \dfrac{1}{n^2}$, which at $x = n$ is $2 + \dfrac{1}{n^2} \ge 2$. So $f_n^2 \not\to x^2$ uniformly. (Same for f_n^3.) It is true if all f_n are bounded.

ANSWER $2f_n$: always uniform. f_n^2, f_n^3: not necessarily (MCQ: option D).

Marks: $2f_n$ 1 · counterexample 2.

13

$K(\varepsilon, x)$ for x/n.

KNOW FIRST

T1 Pointwise convergence M3 The order of quantifiers (a game)

$|x/n - 0| < \varepsilon \iff n > |x|/\varepsilon$. So $K(\varepsilon, x) = \lfloor |x|/\varepsilon \rfloor + 1$ works. Any valid K must exceed $|x|/\varepsilon$ (at $n = K$ we need $K > |x|/\varepsilon$); as $|x| \to \infty$ this grows without bound, so no single K works for all x ∈ ℝ. The convergence is pointwise, not uniform, on ℝ.

Marks: K found 2 · why not uniform 1.

14

$x/(x+n)$ on $[0, a]$ and $[0, \infty)$.

KNOW FIRST

T7 Computing $\|f_n - f\|$ P3 Supremum (least upper bound)

For $x \geq 0$: $0 \leq \dfrac{x}{x+n} \leq \dfrac{x}{n} \to 0$, so the limit is 0. $\dfrac{x}{x+n} = 1 - \dfrac{n}{x+n}$ increases with x. On $[0, a]$: $\|f_n\| = \dfrac{a}{a+n} \to 0$, uniform. On $[0, \infty)$: the values approach 1 as $x \to \infty$ and never reach it, so $\|f_n\| = 1$ for every n: not uniform.

ANSWER $\|f_n\|_{[0,a]} = \dfrac{a}{a+n}$ **(uniform);** $\|f_n\|_{[0,\infty)} = 1$ **(not uniform).**

15

$\arctan(nx) \to \dfrac{\pi}{2}\operatorname{sgn} x$**; not uniform on** $\mathbb{R}$**.**

KNOW FIRST

T2 Finding the limit function T14 Using Theorem 8.2.2

x > 0: $nx \to +\infty$, $\arctan(nx) \to \pi/2$. x < 0: $\to -\pi/2$. x = 0: $\arctan 0 = 0$. So the limit is $\dfrac{\pi}{2}\operatorname{sgn} x$. Each $\arctan(nx)$ is continuous on $\mathbb{R}$, but the limit is discontinuous at 0. If the convergence were uniform, Theorem 8.2.2 would make the limit continuous. Hence not uniform on $\mathbb{R}$.

Marks: limit 1.5 · argument 1.5.

16

$\lim(\cos \pi x)^{2n}$**.**

KNOW FIRST

T2 Finding the limit function P1 Limits of sequences

Let $t = \cos^2 \pi x \in [0, 1]$; the term is t^n. If $x \in \mathbb{Z}$, $\cos \pi x = \pm 1$, $t = 1$, limit 1. Otherwise $0 \le t < 1$, limit 0.

ANSWER The limit exists for all x: $f(x) = 1$ **if x is an integer,** 0 **otherwise.**

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x/n: uniform on [0, 1], not on $\mathbb{R}$.

KNOW FIRST

T9 Uniform depends on the set T4 Proving "NOT uniform"

On [0, 1]: $|x/n| \leq 1/n$, attained at x = 1, so $\|x/n\|_{[0,1]} = 1/n \to 0$: uniform (Lemma 8.1.8). On $\mathbb{R}$: take $\varepsilon_0 = 1$, $x_k = k$: $|f_k(k) - 0| = 1 \geq \varepsilon_0$, so not uniform (Lemma 8.1.5).

18

$xe^{-nx} \to 0$ uniformly on $[0, \infty)$.

KNOW FIRST

T7 Computing $\|f_n - f\|$ M4 Finding a supremum with calculus

Pointwise limit 0 (exponentials beat powers). $\frac{d}{dx}(xe^{-nx}) = e^{-nx}(1 - nx) = 0$ at $x = 1/n$: positive before, negative after, so a maximum. Value $\frac{1}{n}e^{-1}$. At 0 the value is 0 and as $x \to \infty$ it tends to 0.

ANSWER $\|xe^{-nx}\| = \frac{1}{en} \to 0$: **uniform.**

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x^n on $[0, \frac{1}{2}]$.

KNOW FIRST

T9 Uniform depends on the set

x^n increases on $[0, \frac{1}{2}]$, so $\|x^n\|_{[0,1/2]} = (\frac{1}{2})^n = 2^{-n} \to 0$. By Lemma 8.1.8, $x^n \to 0$ uniformly on $[0, \frac{1}{2}]$.

ANSWER $\|x^n\|_{[0,1/2]} = 2^{-n}$.

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Theorem 8.2.2 and its converse.

KNOW FIRST

T13 Uniform limit of continuous functions is continuous **T14** Using Theorem 8.2.2

Theorem. Let (f_n) be a sequence of continuous functions on $A \subseteq \mathbb{R}$ that converges uniformly on A to $f : A \to \mathbb{R}$. Then f is continuous on A.

Converse false. $f_n(x) = x/n$ are continuous on $\mathbb{R}$ and converge to $f = 0$, which is continuous, but the convergence is not uniform on $\mathbb{R}$ ($x_k = k$ gives gap 1). So a continuous limit does not imply uniform convergence.

Marks: statement 1.5 · counterexample 1.5.

Section B · Paragraph / problem
6 marks each.

21 PYQ MAT-21 Q21

Define pointwise convergence; test (x^n).

KNOW FIRST

T1 Pointwise convergence T2 Finding the limit function P1 Limits of sequences

Definition. (f_n) converges on A_0 to f if for each $x \in A_0$, $f_n(x) \to f(x)$ (see Q2).

Test. Let $g_n(x) = x^n$, $x \in \mathbb{R}$.

$x = 1$	$g_n(1) = 1 \to 1$.
$\lvert x\rvert < 1$	$x^n \to 0$.
$x = -1$	$(-1)^n$ oscillates between −1 and 1: divergent.
$\lvert x\rvert > 1$	$\lvert x\rvert^n \to \infty$: unbounded, divergent.

ANSWER (x^n) **converges exactly on** $(-1, 1]$**, to** $g(x) = 0$ **(**$-1 < x < 1$**),** $g(1) = 1$.

The limit is discontinuous at 1 although every x^n is continuous; hence the convergence is not uniform on $(-1, 1]$ (Theorem 8.2.2).

Marks: definition 2 · four cases 3 · conclusion 1.

22 PYQ MAT-21 Q32

Convergence of $x^n(1 - x)$ on [0, 1].

KNOW FIRST

T7 Computing $\|f_n - f\|$ T6 The norm test (Lemma 8.1.8) M4 Finding a supremum with calculus

Pointwise. At $x = 0$ and $x = 1$, $G_n(x) = 0$. For $0 < x < 1$, $x^n \to 0$, so $G_n(x) \to 0$. Limit $G = 0$.

Norm. $G_n'(x) = nx^{n-1}(1-x) - x^n = x^{n-1}\big(n - (n+1)x\big)$, zero at $x_n = \dfrac{n}{n+1}$; $G_n' > 0$ before and < 0 after, so this is the maximum:

$$\|G_n\|_{[0,1]} = \left(\frac{n}{n+1}\right)^n \frac{1}{n+1} = \frac{n^n}{(n+1)^{n+1}} \le \frac{1}{n+1}.$$

(n = 1: ¼; n = 2: 4/27 ≈ 0.148; n = 10: ≈ 0.035.)

ANSWER $\|G_n\| \to 0$, **so** $x^n(1-x) \to 0$ **uniformly on [0, 1].**

Marks: pointwise 1.5 · derivative and critical point 2 · norm 1.5 · conclusion 1.

23 PYQ MAT-22 Q31

Convergence of $\dfrac{x^n}{x^n+1}$ on [0, 2].

KNOW FIRST

T2 Finding the limit function T14 Using Theorem 8.2.2 T13 Uniform limit of continuous functions is continuous

$0 \le x < 1$	$x^n \to 0$, so $f_n(x) \to 0$.
$x = 1$	$f_n(1) = \dfrac{1}{2}$.
$1 < x \le 2$	$f_n(x) = \dfrac{1}{1+x^{-n}} \to 1$ since $x^{-n} \to 0$.

Not uniform on [0, 2]. Each f_n is continuous (quotient of continuous functions, denominator ≥ 1), but the limit f jumps at 1. By Theorem 8.2.2 the convergence cannot be uniform. Directly:

$\sup_{0\le x<1} \dfrac{x^n}{1+x^n} = \dfrac{1}{2}$ (approached as $x \to 1^-$), so $\|f_n - f\| = \dfrac{1}{2}$ for all n.

Where it is uniform. On $[0,b]$, $b<1$: $|f_n| \le x^n \le b^n \to 0$. On $[a,2]$, $a>1$: $|f_n - 1| = \dfrac{1}{1+x^n} \le \dfrac{1}{1+a^n} \to 0$.

ANSWER **Pointwise limit: 0 on [0, 1), ½ at 1, 1 on (1, 2]. Not uniform on [0, 2]; uniform on [0, b] (b < 1) and [a, 2] (a > 1).**

Marks: limit 2 · non-uniformity 2 · uniform subsets 2.

24

Prove Lemma 8.1.5; apply to $(x^2+nx)/n$.

KNOW FIRST

T4 Proving "NOT uniform" M5 Negating a statement T3 Uniform convergence

Lemma. (f_n) does not converge uniformly on A_0 to f iff for some $\varepsilon_0 > 0$ there are a subsequence (f_{n_k}) and points $x_k \in A_0$ with $|f_{n_k}(x_k) - f(x_k)| \ge \varepsilon_0$ for all k.

1. Negating Definition 8.1.4: there is $\varepsilon_0 > 0$ such that for every K there are $n \ge K$ and $x \in A_0$ with $|f_n(x) - f(x)| \ge \varepsilon_0$.
   Flip ∀/∃ and the inequality.

2. K = 1 gives n_1, x_1; $K = n_1 + 1$ gives $n_2 > n_1$, x_2; inductively $n_1 < n_2 < \cdots$: the required subsequence and points.
   Construction by induction.

3 Conversely, if such n_k, x_k exist then for any K some $n_k \geq K$ violates uniformity.

$n_k \to \infty$.

Application. $h_n(x) = \dfrac{x^2 + nx}{n} \to h(x) = x$ (Q6). With $x_k = -k$: $h_k(-k) = \dfrac{k^2 - k^2}{k} = 0$, $h(-k) = -k$, so $|h_k(x_k) - h(x_k)| = k \geq 1 = \varepsilon_0$. Not uniform on $\mathbb{R}$.

Marks: statement 1 · proof 3 · application 2.

25

x^n on [0, 1]: derivatives, limit, why not uniform.

KNOW FIRST

T11 Pointwise limits can break continuity T13 Uniform limit of continuous functions is continuous

$g_n'(x) = nx^{n-1}$ is a polynomial, hence continuous on [0, 1]. Pointwise: $g_n(1) = 1$, and $g_n(x) \to 0$ for $0 \leq x < 1$, so g = 0 on [0, 1), g(1) = 1. Since $\lim_{x \to 1^-} g(x) = 0 \neq 1 = g(1)$, g is not continuous at 1, hence not differentiable at 1 (differentiable ⇒ continuous).

Why not uniform: if $g_n \to g$ uniformly, Theorem 8.2.2 would give g continuous, which is false. (Also $\|g_n - g\| = \sup_{[0,1)} x^n = 1$.)

Marks: derivatives 1 · limit 2 · discontinuity 1 · reason 2.

26

Moving bump $\frac{nx}{1+n^2x^2}$.

KNOW FIRST

T7 Computing $\|f_n - f\|$ T4 Proving "NOT uniform" T9 Uniform depends on the set M4 Finding a supremum with calculus

Pointwise. $f_n(0) = 0$. For $x > 0$: $0 \le f_n(x) \le \frac{nx}{n^2x^2} = \frac{1}{nx} \to 0$. So f = 0.

Norm. $1 + n^2x^2 - 2nx = (1 - nx)^2 \ge 0$, so $f_n(x) \le \frac{1}{2}$, with equality at $x = 1/n$. Hence $\|f_n\|_{[0,\infty)} = \frac{1}{2} \not\to 0$: **not uniform** on $[0, \infty)$.

On $[a, \infty)$. $f_n'(x) = \frac{n(1 - n^2x^2)}{(1 + n^2x^2)^2} < 0$ for $x > 1/n$. For $n > 1/a$, f_n is decreasing on $[a, \infty)$, so $\|f_n\| = f_n(a) = \frac{na}{1 + n^2a^2} < \frac{1}{na} \to 0$: **uniform**.

Marks: pointwise 1.5 · norm ½ 2 · [a, ∞) 2.5.

27

Spikes: $f_n \to 0$, $\int f_n = 1$.

KNOW FIRST

T12 Limits and integrals

$f_n(0) = 0$. For $x \in (0, 1]$: when $n > 2/x$ we have $x > 2/n$, so $f_n(x) = 0$. Hence $f_n(x) \to 0$ for every x: f = 0. The graph on $[0, 2/n]$ is a triangle with base $2/n$ and height n, so $\int_0^1 f_n = \frac{1}{2} \cdot \frac{2}{n} \cdot n = 1$ for all $n \ge 2$.

ANSWER $\lim \int_0^1 f_n = 1 \neq 0 = \int_0^1 \lim f_n$.

This shows that the limit of integrals need not be the integral of the limit for pointwise convergence. Here the limit is continuous, yet the convergence is not uniform: $\|f_n\| = n \to \infty$.

Marks: pointwise 2 · integral 2 · conclusion 2.

28

$(x^2 + nx)/n \to x$ **on [0, 8],** $[-b, b]$**,** $\mathbb{R}$**.**

KNOW FIRST

T9 Uniform depends on the set T7 Computing ‖f_n − f‖ T4 Proving "NOT uniform"

$h_n(x) - x = \dfrac{x^2}{n}$. On [0, 8] its largest value is $\dfrac{64}{n}$ (at x = 8), so $\|h_n - h\|_{[0,8]} = \dfrac{64}{n} \to 0$: uniform. On $[-b, b]$: $\|h_n - h\| = \dfrac{b^2}{n} \to 0$: uniform. On $\mathbb{R}$: x^2/n is unbounded; with $x_k = -k$ the gap is $k \geq 1$ (Q24), so not uniform.

Marks: gap 1 · [0, 8] 2 · [−b, b] 1 · $\mathbb{R}$ 2.

29

$f_n = x + \dfrac{1}{n}$**; squares on** $\mathbb{R}$ **and [0, 1].**

KNOW FIRST

T15 Sums and products T6 The norm test (Lemma 8.1.8)

$|f_n(x) - x| = \dfrac{1}{n}$ for all x, so $f_n \to x$ uniformly on $\mathbb{R}$. $f_n^2(x) - x^2 = \dfrac{2x}{n} + \dfrac{1}{n^2}$. On $\mathbb{R}$, at $x_n = n$ this equals

$2 + \frac{1}{n^2} \geq 2$, so (Lemma 8.1.5 with $\varepsilon_0 = 2$) $f_n^2 \not\to x^2$ uniformly. On [0, 1]: $0 \leq \frac{2x}{n} + \frac{1}{n^2} \leq \frac{2}{n} + \frac{1}{n^2} \to 0$, so uniform. The difference: on [0, 1] all functions are bounded.

Marks: f_n uniform 1 · not on $\mathbb{R}$ 2.5 · on [0, 1] 2.5.

30

$\frac{1}{1+x^n}$ **on [0, 1].**

KNOW FIRST

T14 Using Theorem 8.2.2 T7 Computing $\|f_n - f\|$

P3 Supremum (least upper bound)

For $0 \leq x < 1$: $x^n \to 0$, $f_n(x) \to 1$. At $x = 1$: $f_n(1) = \frac{1}{2}$. So f = 1 on [0, 1), f(1) = ½. The f_n are continuous and f is not, so the convergence is not uniform (Theorem 8.2.2). Norm: for $x < 1$, $|f_n(x) - 1| = \frac{x^n}{1+x^n}$, which increases to ½ as $x \to 1^-$ without reaching it; at x = 1 the gap is 0.

ANSWER $\|f_n - f\|_{[0,1]} = \frac{1}{2}$ **for every n: not uniform.**

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(x^n) **not uniformly Cauchy on [0, 1].**

KNOW FIRST

T10 The Cauchy criterion (Theorem 8.1.10) M4 Finding a supremum with calculus

Each x^n is bounded on [0, 1]. Put $t = x^n \in [0, 1]$: $x^n - x^{2n} = t - t^2 \le \frac{1}{4}$, with equality when $t = \frac{1}{2}$, i.e. $x = 2^{-1/n}$. So $\|f_n - f_{2n}\|_{[0,1]} = \frac{1}{4}$ for every n.

Take $\varepsilon = \frac{1}{8}$. For any H, the choice $m = 2H$, $n = H$ gives $\|f_m - f_n\| = \frac{1}{4} > \varepsilon$. So the Cauchy condition fails and, by Theorem 8.1.10, (x^n) does not converge uniformly on [0, 1] to any bounded function. (The limit function was never used.)

Marks: norm ¼ 3 · failure of Cauchy condition 2 · conclusion 1.

32

$\frac{1}{n}\sin nx$: uniform; derivatives diverge.

KNOW FIRST

T8 The shortcut: a bound free of x **T11** Pointwise limits can break continuity

$|F_n(x)| \le \frac{1}{n}$ for all x, so $\|F_n\|_{\mathbb{R}} \le \frac{1}{n} \to 0$: $F_n \to 0$ uniformly on $\mathbb{R}$. But $F_n'(x) = \cos nx$; at $x = \pi$, $F_n'(\pi) = \cos n\pi = (-1)^n$, which diverges. So (F_n') does not converge (on any set containing π), while the limit F = 0 has $F' = 0$.

This shows: uniform convergence of f_n does not allow differentiating term by term; one needs extra conditions on f_n' (Theorem 8.2.3, practicum).

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x^2e^{-nx} **on** $[0,\infty)$**.**

KNOW FIRST

T7 Computing $\|f_n - f\|$ M4 Finding a supremum with calculus

Pointwise limit 0. $\frac{d}{dx}(x^2e^{-nx}) = xe^{-nx}(2-nx)$: zero at $x = 2/n$, positive before and negative after. Value at 0 and as $x \to \infty$: 0. So

$$\|x^2e^{-nx}\| = \frac{4}{n^2}e^{-2} \approx \frac{0.541}{n^2} \to 0.$$

ANSWER **Uniform on** $[0,\infty)$**; norm** $4/(e^2n^2)$**.**

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$n^2x^2e^{-nx}$ **on** $[0,\infty)$ **and** $[a,\infty)$**.**

KNOW FIRST

T4 Proving "NOT uniform" T9 Uniform depends on the set M4 Finding a supremum with calculus

Pointwise. x = 0: 0. x > 0: $n^2e^{-nx} \to 0$ (exponentials beat powers), times x^2: 0.

Not uniform. The derivative $n^2xe^{-nx}(2-nx)$ vanishes at $x = 2/n$, where the value is $n^2 \cdot \frac{4}{n^2}e^{-2} = \frac{4}{e^2} \approx 0.541$ for **every** n. With $\varepsilon_0 = 4/e^2$ and $x_n = 2/n$, Lemma 8.1.5 shows non-uniformity.

Uniform on $[a,\infty)$**.** For $n > 2/a$ the function decreases on $[a,\infty)$, so the norm is $n^2a^2e^{-na} \to 0$.

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$\frac{nx}{1+nx}$ **on [0, 1]: not uniform, yet integrals converge.**

KNOW FIRST

T12 Limits and integrals T14 Using Theorem 8.2.2
P6 Integrals you can compute

Limit: f(0) = 0, f = 1 on (0, 1] (Q5). The f_n are continuous and f is not, so not uniform (8.2.2); indeed $|f_n(x) - 1| = \frac{1}{1+nx} \to 1$ as $x \to 0^+$, so $\|f_n - f\| = 1$.

Integrals: $\frac{nx}{1+nx} = 1 - \frac{1}{1+nx}$, so

$$\int_0^1 f_n = 1 - \frac{\ln(1+n)}{n} \to 1 = \int_0^1 f$$

(since $\frac{\ln(1+n)}{n} \to 0$; f differs from 1 only at x = 0).

Moral: uniform convergence is sufficient, but not necessary, for $\lim \int f_n = \int f$.

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Sums and products of uniformly convergent sequences.

KNOW FIRST

T15 Sums and products T6 The norm test (Lemma 8.1.8) M1 Inequality kit: $|\cdot|$, ε/2 and ε/3

Sum. For all x: $|(f_n + g_n)(x) - (f+g)(x)| \le |f_n(x) - f(x)| + |g_n(x) - g(x)| \le \|f_n - f\| + \|g_n - g\| \to 0$. So the sum converges uniformly.

Product (all f_n, g_n bounded). Choose N with $\|f_n - f\| \le 1$ for $n \ge N$. Then f is bounded ($|f| \le \|f_N\| + 1$), and so is g; also $\|f_n\| \le \|f\| + 1 =: M$ for $n \ge N$. Now

$$f_n g_n - fg = f_n(g_n - g) + g(f_n - f),$$

so $\|f_n g_n - fg\| \le M\|g_n - g\| + \|g\|\,\|f_n - f\| \to 0$.

Marks: sum 2 · boundedness 2 · product estimate 2.

Section C · Essays

10 marks each.

37 PYQ MTS-25 Q26 also MAT-21 Q38a

State and prove the Cauchy criterion for uniform convergence.

KNOW FIRST

T10 The Cauchy criterion (Theorem 8.1.10) T6 The norm test (Lemma 8.1.8) P4 Cauchy sequences P2 Algebra of limits and the squeeze M1 Inequality kit: $|\cdot|$, ε/2 and ε/3

Theorem 8.1.10. Let (f_n) be a sequence of bounded functions on $A \subseteq \mathbb{R}$. Then (f_n) converges uniformly on A to a bounded function f if and only if for each $\varepsilon > 0$ there is $H(\varepsilon) \in \mathbb{N}$ such that $\|f_m - f_n\|_A \leq \varepsilon$ for all $m, n \geq H(\varepsilon)$.

Proof (⇒). Suppose $f_n \to f$ uniformly. Let $\varepsilon > 0$. By Lemma 8.1.8 there is $K = K(\varepsilon/2)$ with $\|f_n - f\|_A \leq \varepsilon/2$ for $n \geq K$. For $m, n \geq K$ and every $x \in A$,

$|f_m(x) - f_n(x)| \leq |f_m(x) - f(x)| + |f(x) - f_n(x)| \leq$

Taking the supremum over x, $\|f_m - f_n\|_A \leq \varepsilon$ for $m, n \geq H(\varepsilon) := K(\varepsilon/2)$.

Proof (⇐). Suppose the condition holds.

1. Fix $x \in A$. For $m, n \geq H(\varepsilon)$: $|f_m(x) - f_n(x)| \leq \|f_m - f_n\|_A \leq \varepsilon$. So $(f_n(x))$ is a Cauchy sequence in $\mathbb{R}$.
   A value is at most the supremum.

2. By the Cauchy Convergence Criterion (3.5.5), $(f_n(x))$ converges. Define $f(x) := \lim_n f_n(x)$, $x \in A$.
   Completeness of $\mathbb{R}$.

3. Fix $m \geq H(\varepsilon)$. In $|f_m(x) - f_n(x)| \leq \varepsilon$ let $n \to \infty$: $|f_m(x) - f(x)| \leq \varepsilon$ for all $x \in A$.
   Order limit theorem 3.2.6. H does not depend on x.

4. With ε = 1 and $m = H(1)$: $|f(x)| \leq |f_m(x)| + 1 \leq \|f_m\|_A + 1$, so f is bounded.
   f_m is bounded.

5. Thus $\|f_m - f\|_A \leq \varepsilon$ for all $m \geq H(\varepsilon)$, so $f_n \to f$ uniformly on A. ∎
   Lemma 8.1.8.

Marks: statement 2 · (⇒) 3 · (⇐) pointwise Cauchy and definition of f 2 · uniform estimate 2 · bounded 1.

38 PYQ MTS-25 Q20

Uniform limit of continuous functions is continuous; x^n on [0, 1].

KNOW FIRST

T13 Uniform limit of continuous functions is continuous T3 Uniform convergence P5 Continuity at a point (from Module 1) M1 Inequality kit: $|\cdot|$, ε/2 and ε/3 T11 Pointwise limits can break continuity

Theorem 8.2.2. Let (f_n) be a sequence of continuous functions on $A \subseteq \mathbb{R}$ converging uniformly on A to $f : A \to \mathbb{R}$. Then f is continuous on A.

Proof. Let $c \in A$ and $\varepsilon > 0$.

1. By uniform convergence there is $H \in \mathbb{N}$ such that $|f_H(x) - f(x)| < \varepsilon/3$ for all $x \in A$.
   Definition 8.1.4 with ε/3; H independent of x.

2. For $x \in A$: $|f(x) - f(c)| \le |f(x) - f_H(x)| + |f_H(x) - f_H(c)| + |f_H(c) - f(c)| < \dfrac{\varepsilon}{3} + |f_H(x) - f_H(c)| + \dfrac{\varepsilon}{3}$.
   Triangle inequality; step 1 at the two points x and c.

3. Since f_H is continuous at c, there is $\delta > 0$ with $|f_H(x) - f_H(c)| < \varepsilon/3$ whenever $x \in A$, $|x - c| < \delta$.
   Continuity of one function.

4. Hence $|f(x) - f(c)| < \varepsilon$ for $x \in A$, $|x - c| < \delta$. So f is continuous at c, and c was arbitrary. ∎
   Definition 5.1.1.

Deduction. Each x^n is continuous on [0, 1], and $x^n \to g$ pointwise with g = 0 on [0, 1), g(1) = 1, which is discontinuous at 1. If the convergence were uniform, g would be continuous by the theorem. Hence (x^n) does not converge uniformly on [0, 1].

Marks: statement 1 · ε/3 set-up 2 · triangle inequality 2 · δ from continuity of f_H 2 · deduction 3.

39 PYQ MTS-26 Q20 also MAT-22 Q19

Uniform ⇔ $\|f_n - f\|_A \to 0$; apply to x/n.

KNOW FIRST

T6 The norm test (Lemma 8.1.8) T5 Bounded functions and the uniform norm P3 Supremum (least upper bound) T9 Uniform depends on the set

Lemma 8.1.8. A sequence (f_n) of bounded functions on $A \subseteq \mathbb{R}$ converges uniformly on A to f if and only if $\|f_n - f\|_A \to 0$.

(⇒) Let $\varepsilon > 0$. By uniform convergence there is $K(\varepsilon)$ with $|f_n(x) - f(x)| \leq \varepsilon$ for all $x \in A$ and $n \geq K(\varepsilon)$. Then ε is an upper bound of $\{|f_n(x) - f(x)| : x \in A\}$, so its supremum satisfies $\|f_n - f\|_A \leq \varepsilon$ for $n \geq K(\varepsilon)$. As ε is arbitrary, $\|f_n - f\|_A \to 0$.

(⇐) (MAT-22 Q19) Let $\varepsilon > 0$. There is $H(\varepsilon)$ with $\|f_n - f\|_A \leq \varepsilon/2$ for $n \geq H(\varepsilon)$. For every $x \in A$, $|f_n(x) - f(x)| \leq \|f_n - f\|_A \leq \varepsilon/2 < \varepsilon$ when $n \geq H(\varepsilon)$. Since H depends only on ε, $f_n \to f$ uniformly on A. ∎

Application. On [0, 1], $\|x/n - 0\| = \sup_{0 \leq x \leq 1} x/n = 1/n \to 0$: uniform. On $\mathbb{R}$, $f_n - f = x/n$ is **not bounded**, so the norm is not a real number and the lemma does not apply directly. We show non-uniformity directly instead: $x_k = k$ gives $|f_k(k) - 0| = 1$ for all k (Lemma 8.1.5).

Marks: statement 1 · (⇒) 3 · (⇐) 3 · application 3.

40 PYQ MAT-21 Q37b

(x^n) on [0, 1], $[0, b]$, [0, 1).

KNOW FIRST

T9 Uniform depends on the set T4 Proving "NOT uniform" T7 Computing $\|f_n - f\|$ T14 Using Theorem 8.2.2

Limit. $g(x) = 0$ for $0 \le x < 1$, $g(1) = 1$.

On [0, 1]: not uniform. Three independent proofs:

1. Points: $x_k = 2^{-1/k}$ gives $|x_k^k - g(x_k)| = \frac{1}{2}$ for every k (Lemma 8.1.5, $\varepsilon_0 = ½$).
2. Norm: $\|x^n - g\|_{[0,1]} = \sup_{0 \le x < 1} x^n = 1 \not\to 0$ (Lemma 8.1.8).
3. Continuity: the x^n are continuous, g is not (Theorem 8.2.2).

On $[0, b]$, $0 < b < 1$: uniform. x^n increases in x, so $\|x^n - 0\|_{[0,b]} = b^n \to 0$.

On [0, 1): not uniform, even though the limit 0 is continuous there: $\sup_{0 \le x < 1} x^n = 1$ for every n (proof 1 also works, as $x_k \in [0, 1)$).

ANSWER Not uniform on [0, 1] or [0, 1); uniform on every [0, b] with b < 1.

Marks: limit 2 · [0, 1] 3 · [0, b] 2 · [0, 1) with remark 3.